

STOCK VOLATILITY MODELING

HOMEWORK TEAM 2

**CARLOS ALONZO-MONTUFAR, ALI AWAN,
DELANEY BLACK, LIAM CAWLEY, AND PRIYA
CHILUKURI**

FRIDAY, FEBRUARY 27, 2026

Table of Contents

Overview	1
Methodology	1
Data Used	1
Stock Selection	1
Model Development & Forecasting	1
Analysis	1
Results	2
Recommendations	4
Conclusion	4

STOCK VOLATILITY MODELING

Overview

The Commercial Banking Corporation (the “Bank”) aims to evaluate the volatility of the thirty stocks present in the Dow Jones Industrial Average (DIJA). Our team collected daily stock prices and returns for the DIJA from March 1, 2011, to March 24, 2021. We implemented ARCH/GARCH modeling to predict the most significant stocks in the DIJA and measure sensitivity and persistence. Out of the five most significant stocks, we recommend investing in Honeywell for a lower-risk portfolio, Microsoft for a middle-risk portfolio, and Travelers, Goldman Sachs, and Boeing for a higher-risk, higher-reward portfolio.

Methodology

This section discusses the methodology used to select the most significant stocks in the DIJA and develop an ARCH/GARCH model to predict volatility.

Data Used

We first identified the stocks currently present in the DIJA and their tickers using Wikipedia, accurate as of February 24, 2026. For each stock, we collected daily stock prices and returns from March 1, 2011, to March 24, 2021, using the yfinance package in Python.

Stock Selection

We ranked the thirty stocks from the most to least significant using the Lagrange Multiplier Lag 1 test p-values. Using this ranking, we identified the five most significant stocks for our subsequent ARCH/GARCH models. These stocks are Travelers, Boeing, Honeywell, Goldman Sachs, and Microsoft.

Model Development & Forecasting

We tested four ARCH/GARCH models for each of the stocks identified above to predict volatility. These models include a GARCH(1,1)-Normal, GARCH(1,1)-t, GJR-GARCH(1,1,1)-Normal, and GJR-GARCH(1,1,1)-t. We opted to use the GJR-GARCH model rather than the QGARCH model due to limitations in the arch Python package. For all five stocks, GJR-GARCH(1,1,1)-t yielded the lowest BIC, so we opted to use it for our following analysis. After selecting our preferred model, we forecasted the next 30 days of volatility for each stock. We then ranked these stocks first by those most likely to be affected by a shock to the market today, and then by those we expect to experience the longest effect after a shock.

Analysis

After fitting the models, we found that more sensitive stocks tended to have shorter persistence times, while less sensitive stocks tended to have longer persistence times. However, Goldman Sachs is both relatively sensitive to shocks and highly persistent in volatility. It's possible that Goldman Sachs investors are somewhat risk-averse and will sell off Goldman Sachs when a negative shock happens. It's also possible that they remain wary for a while after.

For the top five stocks, the GARCH(1,1)-normal model yielded persistence values greater than .85 but less than one, and half-lives ranging from about four days to 25 days, indicating that stock “memory” can be long-lived, extending for almost a month after the shock. All five stocks had alpha values near 0.10, suggesting at least some short-term sensitivity to shocks.

By contrast, the GARCH(1,1)-t model calculated higher persistence values for each of the top five stocks. This behavior is consistent with the t-distribution’s fatter tails compared to a normal distribution, allowing for more extreme events than would be predicted by a normal distribution. The GJR-GARCH(1,1,1)-t model had the lowest BIC, suggesting the best fit among the models we tried. It’s reasonable that GJR-GARCH fits the data better than a standard GARCH model, given that negative shocks intuitively are more impactful than positive shocks.

GJR-GARCH gave persistence values similar to those from the GARCH(1,1)-t model. However, there were significant disparities in the calculated leverage values: Honeywell’s leverage was nearly 20 times that of the next-highest, suggesting that Honeywell is uniquely susceptible to bad news compared to the other stocks. Given that Honeywell is a multi-industry conglomerate with relatively low sensitivity compared to other stocks, it makes sense that shocks would rarely affect Honeywell, but those that do would affect the market as a whole, making them very intense. Of the top five stocks, the minimum leverage value was 2.30, suggesting that bad news is at least twice as negatively impactful as good news is positively impactful.

Results

The following results evaluate the GJR-GARCH(1,1,1)-t model's ability to forecast the volatility of the top five stock components, ranking them by sensitivity and persistence.

According to our model, the top five most significant stock components, ranked by the LM test, are Travelers, Honeywell, Microsoft, Boeing, and Goldman Sachs. Table 1 illustrates the sensitivity, persistence, and half-life of the top five stock components.

Table 1: Stock component comparative analysis

Stock Component	Sensitivity Ranking	Persistence Ranking	Alpha	Half-Life Days	Leverage Ratio
Travelers	1	5	0.08	8.36	2.28x
Honeywell	5	1	0.00	27.54	104.20x
Goldman Sachs	2	2	0.05	22.38	2.28x
Microsoft	3	4	0.04	13.19	5.18x
Boeing	4	3	0.03	20.68	3.44x

Based on the sensitivity ranking, the stocks most likely to be affected by a market shock today, in descending order, are Travelers, Goldman Sachs, Microsoft, Boeing, and Honeywell. Based on the persistence ranking, the stocks with the longest-lasting effects of a shock, in descending order, are Honeywell (HON), Goldman Sachs (GS), Boeing (BA), Microsoft (MSFT), and Travelers (TRV). The leverage ratio displays the effect of a negative shock on a stock. Honeywell's negative shocks hit 104.20x harder than its positive shocks. Figure 1 illustrates the comparison between the sensitivity and the half-life of the top five stocks.

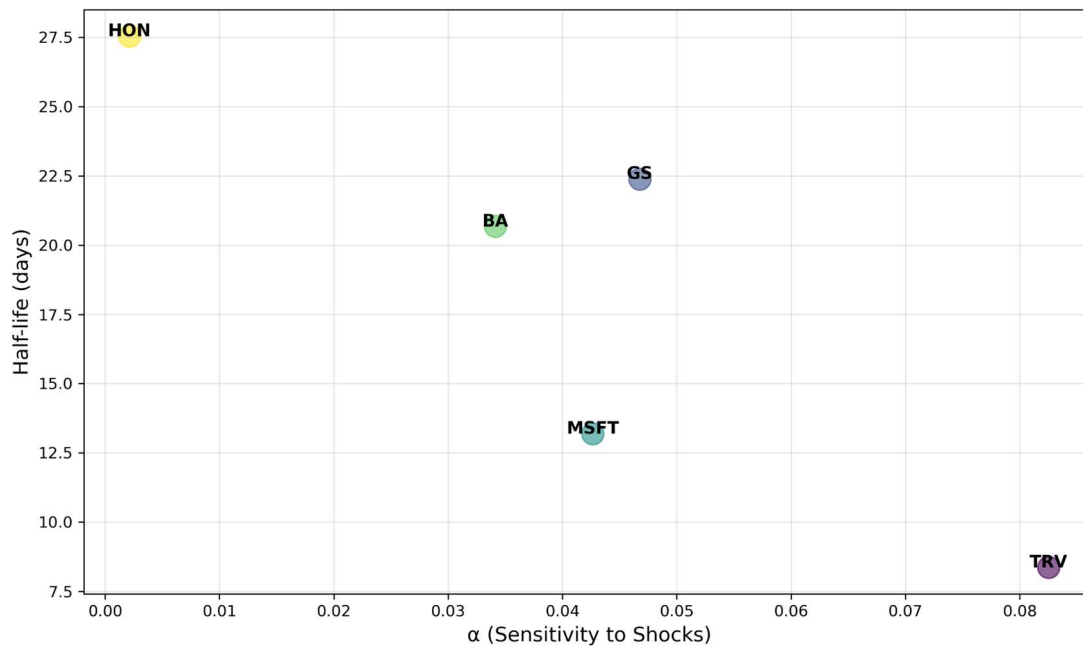


Figure 1: Shock Sensitivity vs Persistence

Honeywell is very clearly the most persistent to long-term shocks, but very insensitive to short-term shocks, whereas Travelers is the opposite. Microsoft, Boeing, and Goldman Sachs are roughly balanced between sensitivity and persistence, with Microsoft being the least persistent of the three. Using our model, we also produced a 30-day volatility forecast for each of the top five stocks, which is illustrated in Figure 2.

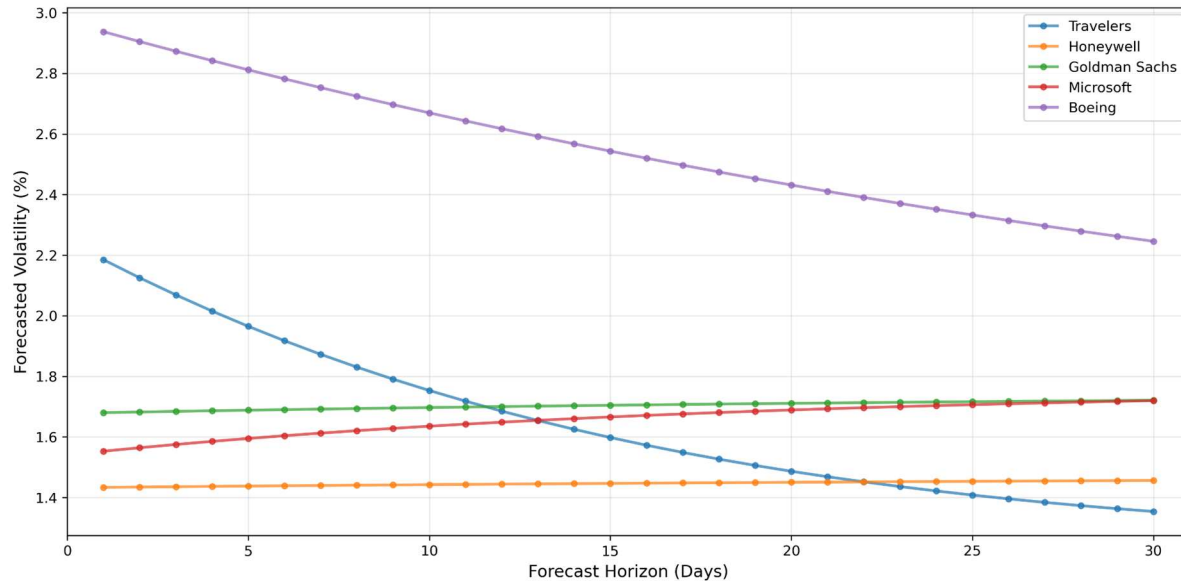


Figure 2: 30-day volatility forecast for the top five stocks

Travelers and Boeing's volatility are expected to decrease over the next 30 days, whereas Goldman Sachs, Honeywell, and Microsoft's volatility is expected to increase.

Recommendations

This section presents recommendations for the Bank's Portfolio Pricing department and its portfolio management.

We recommend that the Bank use the GJR-GARCH(1,1,1)-t model for ongoing monitoring, as it consistently minimizes the BIC. The Bank should also closely monitor Goldman Sachs, Honeywell, and Microsoft due to their rising volatility and align portfolios with their appropriate risk tolerances. Microsoft appears to be a relatively safe stock due to its balance between sensitivity and persistence; however, with the predicted increase in volatility, this should be approached with caution.

Unless the Bank has a higher risk appetite, we recommend avoiding Travelers' stock because it is very sensitive to short-term shocks. If the Bank has a low appetite for risk, we recommend allocating capital to Honeywell stocks due to their high persistence and incredibly low sensitivity, with the caveat of close monitoring. Honeywell's 104.20x leverage indicates that negative shocks have a significant impact on the stock's volatility, which could cause serious repercussions should such an event occur. If the Bank desires a more balanced portfolio, we recommend Microsoft stocks for their balance between persistence and sensitivity.

Conclusion

Our team recommends a GJR-GARCH(1,1,1)-t model for ongoing monitoring of the stocks present in the DIJA. We have identified the five most significant stocks: Travelers, Honeywell, Goldman Sachs, Microsoft, and Boeing. For portfolios with a high-risk appetite, we recommend

Travelers, Boeing, and Goldman Sachs due to their high sensitivity and lower persistence. For portfolios with a low-risk appetite, we recommend focusing on Honeywell stocks due to their high persistence and low sensitivity. For balanced portfolios, we recommend Microsoft stocks due to their balance between sensitivity and persistence. We have also forecast volatility over the next 30 days for each of these stocks, indicating that the Bank should closely monitor portfolios that include Honeywell, Goldman Sachs, and Microsoft due to the expected increase in volatility. Utilizing our model and adjusting portfolio prices based on the desired risk appetite and our recommendations will enable the Bank's Portfolio Pricing department to make more informed decisions regarding its services.